

St George Girls High School

Year 12

Mid-HSC Course Examination

2006



Mathematics

Extension 1

General Instructions

- Working time – 1½ hours
- Reading time – 5 minutes
- Write using blue or black pen
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.
- Write on one side of the page only.
- Start each question in a **new booklet**.

Total marks – 75

- Attempt Questions 1 – 5
- All questions are of equal value

Question 1 – (15 marks) – Start a new booklet

Marks

a) Differentiate with respect to x

(i) $y = e^{\sin x}$ 1

(ii) $y = \log_e \left(\frac{\sqrt{x^2 + 1}}{x + 3} \right)$ 2

b) (i) Write down the domain of the function $\ln(4 - x)$ 1

(ii) Draw a neat sketch of the curve $y = \ln(4 - x)$ 2

c) (i) Find $\frac{d}{dx}(x^3 \ln x)$ 1

(ii) Hence, or otherwise, find the value of $\int_1^e x^2 \ln x \, dx$ 3

d) The value, $\$V$, of a car depreciated with time such that $\frac{dV}{dt} = -kV$ for some constant $k > 0$.

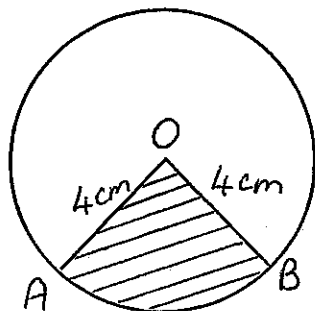
(i) Show that $V = Ae^{-kt}$ satisfies the differential equation. 1

(ii) After 2 years the value is \$19 808 and after a further 5 years the value is \$5135. Find the values of k and A . 4

Question 2 – (15 marks) – Start a new booklet

Marks

a)



The area of minor sector AOB is 20cm^2 .

(i) Find the size of $\hat{AOB}$. 2

(ii) Hence find the length of major arc AB . 2

b) Find the exact value of

(i) $\tan \frac{2\pi}{3}$ 1

(ii) $\cos 105^\circ$ 2

c) Solve for $0 \leq x \leq 2\pi$

(i) $\operatorname{cosec} \left(x - \frac{\pi}{4} \right) = -\sqrt{2}$ 2

(ii) $\sqrt{3} \cos x = 2 \sin x \cos x$ 3

(iii) $\tan^2 x - \sec x - 1 = 0$ 3

Question 3 – (15 marks) – Start a new booklet

Marks

- a) Find the equation of the tangent to the curve $y = \cos 2x$ at the point where

$$x = \frac{\pi}{6}$$

3

- b) Find the following:

(i) $\int \sin 3x \, dx$

1

(ii) $\int \sin^2 3x \, dx$

2

(iii) $\int \cos^4 x \sin x \, dx$

2

(iv) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan x \, dx$

3

- c) (i) Show that the line $y = 4 - 2x$ crosses $y = \ln(x - 1)$ at the point $(2, 0)$ 1

- (ii) Find the acute angle between $y = 4 - 2x$ and $y = \ln(x - 1)$ at the point $(2, 0)$ 3

Question 4 – (15 marks) – Start a new booklet

Marks

- a) (i) By considering the derivative, or otherwise, show that $f(x) = \frac{e^x}{1+e^x}$ is an increasing function. 2
- (ii) Explain why the inverse of $f(x)$ is a function. 1
- (iii) Find $f^{-1}(x)$, this inverse function. 3
- b) (i) Write down the domain and range of $y = 3\cos^{-1}(x+1)$ 2
- (ii) Draw a neat sketch of the graph of $y = 3\cos^{-1}(x+1)$ 2
- c) Find the exact value of $\tan\left(2\sin^{-1}\left(\frac{3}{4}\right)\right)$ 3
- d) Show that $\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 5x} = \frac{3}{5}$ 2

Question 5 – (15 marks) – Start a new booklet

Marks

a) Find the following:

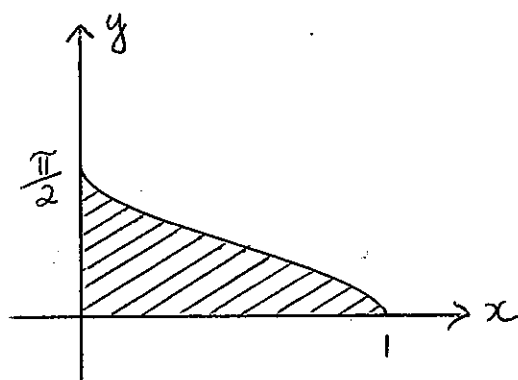
(i) $\int \frac{dx}{\sqrt{25-4x^2}}$ 2

(ii) $\int \frac{dx}{16+9x^2}$ 2

b) If $\sin x = \cos \frac{\pi}{10}$ find all possible values of x .

3

c) A sketch of the graph of $y = \cos^{-1} \sqrt{x}$ is shown below.



Find the shaded area.

3

(HINT: Find the area between the curve and the y axis)

d) (i) Show that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} (-1 \leq x \leq 1)$

2

(ii) If $\sin^{-1} A + \cos^{-1} B = \frac{11\pi}{12}$ and $\cos^{-1} A - \sin^{-1} B = \frac{5\pi}{12}$ find A and B .

3

(HINT: Use the result from part (i))

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

QUESTION 1

(a) (i) $\frac{d}{dx} [e^{\sin x}] = \cos x \cdot e^{\sin x}$

(ii) $\frac{d}{dx} [\log_e (x^2+1)^{\frac{1}{2}} - \log_e (x+3)]$

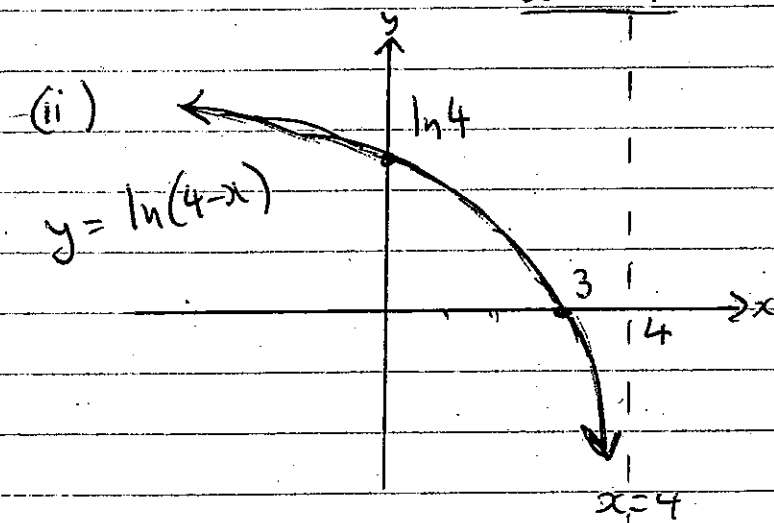
$$= \frac{d}{dx} \left[\frac{1}{2} \log_e (x^2+1) - \log_e (x+3) \right]$$

$$= \frac{1}{2} \times \frac{2x}{x^2+1} - \frac{1}{x+3}$$

$$= \frac{x}{(x^2+1)} - \frac{1}{(x+3)}$$

$$\begin{aligned} & * \frac{x^2+3x-x^2+1}{(x^2+1)(x+3)} \\ & = \frac{3x+1}{(x^2+1)(x+3)} \end{aligned}$$

(b) (i) Domain: $4-x > 0$
 $-x > -4$
 $x < 4$



(c) (i) $\frac{d}{dx} (x^3 \ln x) = \ln x \cdot 3x^2 + x^3 \cdot \frac{1}{x}$

$$= 3x^2 \ln x + x^2$$

$$(ii) \text{ From (i) } 3x^2 \ln x = \frac{d}{dx} (x^3 \ln x) - x^2$$

$$\text{so } x^2 \ln x = \frac{1}{3} \left[\frac{d}{dx} (x^3 \ln x) - x^2 \right]$$

$$\begin{aligned} \int_1^e x^2 \ln x \, dx &= \frac{1}{3} \int_1^e \left[\frac{d}{dx} (x^3 \ln x) - x^2 \right] dx \\ &= \frac{1}{3} \left[x^3 \ln x - \frac{x^3}{3} \right]_1^e \\ &= \frac{1}{3} \left[(e^3 \times 1 - \frac{e^3}{3}) - (0 - \frac{1}{3}) \right] \\ &= \frac{1}{3} \left[e^3 - \frac{e^3}{3} + \frac{1}{3} \right] \\ &= \frac{2}{9} e^3 + \frac{1}{9} \end{aligned}$$

$$\begin{aligned} (d) \quad (i) \quad \frac{dV}{dt} &= \frac{d}{dt} [A e^{-kt}] \\ &= -k \cdot A e^{-kt} \\ &= -kV \quad \text{as required.} \end{aligned}$$

$$(ii) \quad t=2, \quad V=19808 \quad \text{and} \quad t=7, \quad V=5135$$

$$19808 = A e^{-2k} \quad \dots (1) \quad \quad 5135 = A e^{-7k} \quad \dots (2)$$

$$(1) \div (2) \quad \frac{19808}{5135} = \frac{A e^{-2k}}{A e^{-7k}}$$

$$\therefore e^{5k} = 3.857 \dots$$

$$5k = \ln(3.857 \dots)$$

$$k = \frac{1}{5} \ln(3.857 \dots)$$

$$k = 0.27 \quad (2 \text{ dec. pl.})$$

$$\text{So } \frac{19808}{e^{-0.54}} = A$$

$$\text{Gives } A = 33991 \quad (\text{nearest whole \#})$$

QUESTION 2

(a) (i) let $\hat{AOB} = \theta$
 then $A = \frac{1}{2} r^2 \theta$
 $20 = \frac{1}{2} \times 4^2 \times \theta$
 $\frac{5}{2} \text{ rad} = \theta$

(ii) Arc length $l = r\theta$ of minor arc
 $= 4 \times \frac{5}{2}$
 $= 10 \text{ cm}$

$\therefore$ Major arc $2\pi \times 4 - 10$
 $= (8\pi - 10) \text{ cm}$
 $= 15.1 \text{ cm}$

(b) (i) $\tan \frac{2\pi}{3} = -\tan \frac{\pi}{3}$
 $= -\sqrt{3}$

(ii) $\cos 105^\circ = \cos (60^\circ + 45^\circ)$
 $= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$
 $= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$
 $= \frac{1 - \sqrt{3}}{2\sqrt{2}}$

(c) (i) $\sin \left(x - \frac{\pi}{4} \right) = -\frac{1}{\sqrt{2}}$

$$0 \leq x \leq 2\pi$$

$$-\frac{\pi}{4} \leq x - \frac{\pi}{4} \leq 2\pi - \frac{\pi}{4}$$

$$-\frac{\pi}{4} \leq x - \frac{\pi}{4} \leq \frac{7\pi}{4}$$

so $x - \frac{\pi}{4} = \frac{\pi}{4}, \left(\pi + \frac{\pi}{4} \right), \left(2\pi - \frac{\pi}{4} \right)$
 $\frac{\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$
 $x = 0, \frac{6\pi}{4}, \frac{8\pi}{4}$
 $= 0, \frac{3\pi}{2}, 2\pi$

(ii) $\sqrt{3} \cos x - 2 \sin x \cos x = 0$
 $\cos x (\sqrt{3} - 2 \sin x) = 0$

$$\text{let } \cos x = 0 \quad \text{and} \quad \sqrt{3} - 2 \sin x = 0$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$2 \sin x = \sqrt{3}$$

$$\sin x = \frac{\sqrt{3}}{2}$$

$$x = \frac{\pi}{3}, \pi - \frac{\pi}{3}$$

$$= \frac{\pi}{3}, \frac{2\pi}{3}$$

$$(iii) \tan^2 x = \sec^2 x - 1 \quad \underline{\text{so}} \quad \tan^2 x - \sec x - 1 = 0$$

$$\text{becomes: } \sec^2 x - \sec x - 2 = 0$$

$$(\sec x - 2)(\sec x + 1) = 0$$

$$\text{let } \sec x = 2 \quad \text{and} \quad \sec x = -1$$

$$\cos x = \frac{1}{2}$$

$$\cos x = -1$$

$$x = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, \pi$$

$$= \frac{\pi}{3}, \frac{5\pi}{3}, \pi$$

QUESTION 3

(a) $\frac{dy}{dx} = -2 \sin 2x$ when $x = \frac{\pi}{6}$, $y = \cos \frac{\pi}{3} = \frac{1}{2}$
and gradient $m = -2 \sin \frac{\pi}{3} = -\sqrt{3}$

Equation of tangent

$$y - \frac{1}{2} = -\sqrt{3} \left(x - \frac{\pi}{6} \right)$$

$$y = -\sqrt{3} \cdot x + \frac{\pi\sqrt{3}}{6} + \frac{1}{2}$$

(b) (i) $\int \sin 3x \cdot dx = -\frac{1}{3} \cos 3x + C$

(ii) $\int \sin^2 3x \cdot dx = \int \frac{1}{2} (1 - \cos 6x) \cdot dx$
 $= \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$
 $= \frac{x}{2} - \frac{\sin 6x}{12} + C$

(iii) $\int \cos^4 x \cdot \sin x \cdot dx = - \int (-\sin x) (\cos x)^4 dx$
 $= - \frac{(\cos x)^5}{5} + C$
 $= - \frac{\cos^5 x}{5} + C$

(iv) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan x \cdot dx = - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sin x}{\cos x} \cdot dx$
 $= - \left[\ln(\cos x) \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$
 $= - \left[\ln\left(\cos \frac{\pi}{3}\right) - \ln\left(\cos \frac{\pi}{4}\right) \right]$
 $= - \left(\ln\left(\frac{1}{2}\right) - \ln\left(\frac{1}{\sqrt{2}}\right) \right)$
 $= \ln\left(\frac{1}{\sqrt{2}}\right) - \ln\left(\frac{1}{2}\right)$
 $= \ln\left(\frac{2}{\sqrt{2}}\right)$
 $= \ln(\sqrt{2})$

$= \frac{1}{2} \ln 2$

(c) (i) $y = 4 - 2x$
 when $x = 2$
 $y = 0$

$y = \ln(x-1)$
 when $x = 2$
 $y = \ln 1$
 $= 0$

$\therefore (2, 0)$ lies on both curves.

(ii) $y' = -2$

$y' = \frac{1}{x-1}$

$m_1 = -2$

at $x = 2$

$m_2 = \frac{1}{2-1}$

$= 1$

If θ is acute angle between curves at $(2, 0)$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-2 - 1}{1 + (-2) \times 1} \right|$$

$$= 3$$

$\therefore \theta = 71^\circ 34' \text{ (nearest min)}$

QUESTION 4:

$$(a) (i) f'(x) = \frac{(1+e^x)e^x - e^x \cdot e^x}{(1+e^x)^2}$$

$$= \frac{e^x + e^{2x} - e^{2x}}{(1+e^x)^2}$$

$$= \frac{e^x}{(1+e^x)^2}$$

Since $e^x > 0$ for all x

and $(1+e^x)^2 > 0$ for all x

then $f'(x) > 0$ for all x and function is increasing over domain.

(ii) Since $f(x)$ increasing for all x any horizontal line only cuts the curve in one point. It follows that the inverse will pass vertical line test; hence, the inverse will be a function.

That is, one-to-one correspondence for x and y values.

$$(iii) \quad y = \frac{e^x}{1+e^x}$$

$$\text{then inverse } x = \frac{e^y}{1+e^y}$$

$$x + x \cdot e^y = e^y$$

$$(x-1)e^y = -x$$

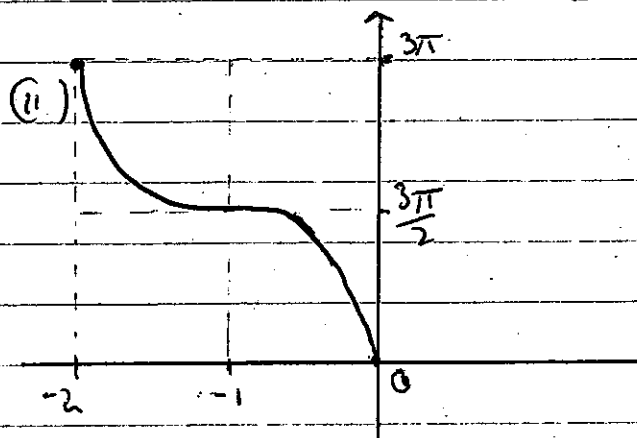
$$e^y = \frac{x}{1-x}$$

$$\text{th } y = \ln \left(\frac{x}{1-x} \right)$$

$$\text{Gives } f^{-1}(x) = \ln \left(\frac{x}{1-x} \right)$$

(b) (i) Domain : $-1 \leq x+1 \leq 1$
 $-2 \leq x \leq 0$

Range : $0 \leq \frac{y}{3} \leq \pi$ $\left[0 \leq \cos^{-1} f(x) \leq \pi \right]$
 $0 \leq y \leq 3\pi$ $\therefore 0 \leq 3\cos^{-1} f(x) \leq 3\pi$



(c) $\tan \left(2 \sin^{-1} \left(\frac{3}{4} \right) \right)$

let $\sin^{-1} \left(\frac{3}{4} \right) = \alpha$

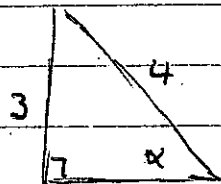
then $\frac{3}{4} = \sin \alpha$

Now $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$

$= \frac{2 \times \frac{3}{\sqrt{7}}}{1 - \left(\frac{3}{\sqrt{7}} \right)^2}$

$= \frac{\frac{6}{\sqrt{7}} \times \frac{-7}{2}}{1 - \frac{9}{7}}$

$= -3\sqrt{7}$



$= \sqrt{7}$

So $\tan \alpha = \frac{3}{\sqrt{7}}$

(d) $\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 5x} = \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \times \frac{5x}{\tan 5x} \right) \times \frac{3}{5}$

$= \frac{3}{5} \times \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \lim_{x \rightarrow 0} \frac{5x}{\tan 5x}$

$= \frac{3}{5} \times 1 \times 1$

$= \frac{3}{5}$

Question 5:

(i) $\int \frac{dx}{2\sqrt{\frac{25}{4}-x^2}} = \frac{1}{2} \sin^{-1} \frac{2x}{5} + C \quad \text{or} \quad -\frac{1}{2} \cos^{-1} \frac{2x}{5} + C$

[use $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a}$]

(ii) $\int \frac{dx}{9(\frac{16}{9}+x^2)} = \frac{1}{9} \times \frac{1}{\frac{4}{3}} \times \tan^{-1} \frac{x}{\frac{4}{3}} + C$
 $= \frac{1}{12} \tan^{-1} \frac{3x}{4} + C$

(3) Now $\cos\left(\frac{\pi}{10}\right) = \sin\left(\frac{\pi}{2} - \frac{\pi}{10}\right)$

$\therefore \sin x = \sin\left(\frac{\pi}{2} - \frac{\pi}{10}\right)$

$\sin x = \sin \frac{2\pi}{5}$

$x = \frac{2\pi}{5}, \pi - \frac{2\pi}{5}, 2\pi + \frac{2\pi}{5}, \dots \quad n \in \mathbb{Z}$

$= k\pi + (-1)^k \frac{2\pi}{5} \quad k \in \mathbb{Z}$

(4) $y = \cos^{-1} \sqrt{x}$

$\cos y = \sqrt{x}$

$\cos^2 y = x$

So Area bounded by curve and y-axis

$A = \int_0^{\frac{\pi}{2}} (\cos^2 y) dy$

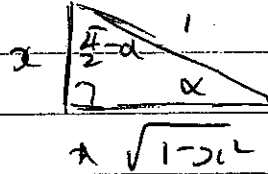
$= \int_0^{\frac{\pi}{2}} \frac{1}{2} [\cos 2y + 1] dy$

$= \frac{1}{2} \left[\frac{1}{2} \sin 2y + y \right]_0^{\frac{\pi}{2}}$

$= \frac{1}{2} \left[\left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin 0 + 0 \right) \right]$

$= \frac{\pi}{4} \text{ sq. units.}$

(d) (i) let $\sin^{-1} x = \alpha$
 $x = \sin \alpha$



then $\alpha = \cos\left(\frac{\pi}{2} - \alpha\right)$

So

$$\cos^{-1} x = \frac{\pi}{2} - \alpha$$

Thus $\sin^{-1} x + \cos^{-1} x = \alpha + \frac{\pi}{2} - \alpha$
 $= \frac{\pi}{2}$

or $\frac{d}{dx} [\sin^{-1} x + \cos^{-1} x] = \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}}$
 $= 0$

$\therefore \sin^{-1} x + \cos^{-1} x = C$ (C is constant)

Subst $x = \frac{1}{2}$

$$\begin{aligned} y &= \sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) \\ &= \frac{\pi}{6} + \frac{\pi}{3} \\ &= \frac{3\pi}{6} \\ &= \frac{\pi}{2} \end{aligned}$$

$$C = \frac{\pi}{2}$$

(ii) From (i) $\sin^{-1} A = \frac{\pi}{2} - \cos^{-1} A$
 $\cos^{-1} B = \frac{\pi}{2} - \sin^{-1} B$

So $\sin^{-1} A + \cos^{-1} B = \frac{11\pi}{12} \quad \dots \textcircled{*}$

is $\left(\frac{\pi}{2} - \cos^{-1} A\right) + \left(\frac{\pi}{2} - \sin^{-1} B\right) = \frac{11\pi}{12}$
 $= \cos^{-1} A - \sin^{-1} B = -\frac{\pi}{12}$

$\cos^{-1} A + \sin^{-1} B = \frac{\pi}{12} \quad \dots \textcircled{1}$

$$\cos^{-1} A - \sin^{-1} B = \frac{5\pi}{12} \quad \dots (2)$$

$$\cos^{-1} A + \sin^{-1} B = \frac{\pi}{12}$$

①+②

$$2\cos^{-1} A = \frac{\pi}{2}$$

$$\cos^{-1} A = \frac{\pi}{4}$$

$$\therefore A = \cos \frac{\pi}{4}$$

$$A = \frac{1}{\sqrt{2}}$$

①-②

$$-2\sin^{-1} B = \frac{\pi}{3}$$

$$\sin^{-1} B = -\frac{\pi}{6}$$

$$B = \sin\left(-\frac{\pi}{6}\right)$$

$$= -\sin \frac{\pi}{6}$$

$$= -\frac{1}{2}$$